

Refining the Sunspot Number Series to SN V3

RECONSTRUCTING A PROVENANCE-RICH SOLAR ACTIVITY RECORD

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We will cover

- Why the Sunspot Number needs a ground-up reconstruction.
- FARSuN's four-axis workflow and quality gates.
- Highlights from key historical datasets driving SN V3.
- Expected benefits, community roadmap, and ways to engage.

Reference material

- FARSuN interactive report ([webpage.html](#)).
- ESWW25 posters: FARSuN Reconstruction & SN Next-Gen.
- Dataset dossiers (QMD files) for Zürich, Mittheilungen, Adams, Gruithuisen, Stark.

Mission Brief

Sunspot Number Next-Gen

Why FARSuN?

The Sunspot Number (SN, formerly the International Sunspot Number) underpins space climate diagnostics, yet two major recalibrations still leave observer-dependent jumps (e.g. the 1947 "Waldmeier break").

FARSuN (Findability and Accessibility of historical Raw Sunspot Numbers) is rebuilding the index from primary sources (1607–1980) with traceable metadata, uncertainties, and open delivery.

Mission Recover and standardise dispersed sunspot archives (drawings, tables, weather notes).

Outcome Deliver SN V3 with hemispheric indices, provenance, and FAIR-compliant access.

Project Pulse (Early 2025)

1607–1980

Temporal coverage of curated sources

95% of 2 000+

Zürich observation tables digitised

(Transkribus + manual QC)

1 056

Entries from C. H. Adams notebooks

with spotless-day tagging

100%

Mittheilungen volumes integrated

into database

400+

Unique historical observers linked to

standard IDs

Ready for pilot

Citizen-science portal blueprint and

QA workflow

Source: FARsUN interactive report and dataset dossiers (Zürich, Mittheilungen, Adams, Gruithuisen, Stark).

Why Reconstruction

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Limitations of Legacy Recalibration

Symptoms

- Residual scale jumps ("Waldmeier jump", late-19th c. transitions) in SN V2 derived records.
- Missing metadata: observer aliases, instrument notes, weather context drop out of composite series.
- Uncertainties opaque; difficult to quantify spotless-day reliability or observer drift.

Why Tweaking Coefficients Is Not Enough

- Linear rescaling keeps biases embedded in the stitched index.
- Downstream models inherit hidden assumptions, limiting predictive skill.
- Science questions (hemispheric asymmetry, climate forcing) demand provenance-rich data.

Reconstruction Advantages

Legacy recalibration (SN V2)

- Adjusts existing composite, no new metadata.
- Biases from observer transitions persist.
- Limited uncertainty audit trail.

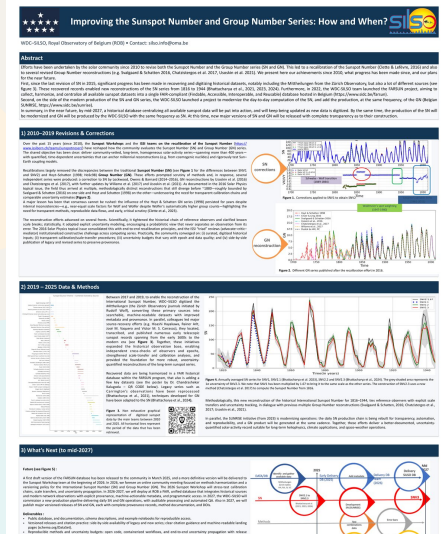
FARSuN reconstruction (SN V3)

- Revisits original artefacts (drawings, logbooks, weather notes).
- Builds calibration curves per observer using raw counts.
- Publishes machine-readable products with explicit uncertainties and provenance.

Narrative adapted from the FARSuN interactive report (webpage.html).

Coordinating with the SN Next-Gen Programme

- The SN Next-Gen (SNGN) roadmap aligns international partners on governance, release cadence, and downstream services for the rebuilt record.
- FARSuN supplies the data backbone: curated sources, calibration curves, and uncertainty models feeding SN V3.
- Joint milestones for ESWW25:
 - Harmonised metadata schema (observer registry, instrument vocabularies).
 - Shared validation dashboards comparing historical reconstructions and modern proxies.
- Communication kit for agencies and

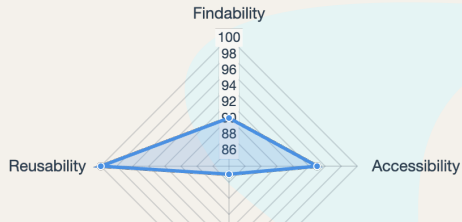


Methodology

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Four-Axis Workflow

1. **Gather** Locate and digitise dispersed historical sources (Wolf's *Mittheilungen*, Zürich tables, Gruithuisen ledgers, Adams drawings).
2. **Process** Blend handwriting/OCR engines (Transkribus) with manual keying for fragile tables and drawings.
3. **Validate** Reconcile observer IDs, calendars, and counting conventions; propagate spotless-day confidence levels.
4. **Disseminate** Release FAIR-ready datasets, APIs, and VO-compliant services with versioned uncertainties.



Quality Gates and Tooling

Verification

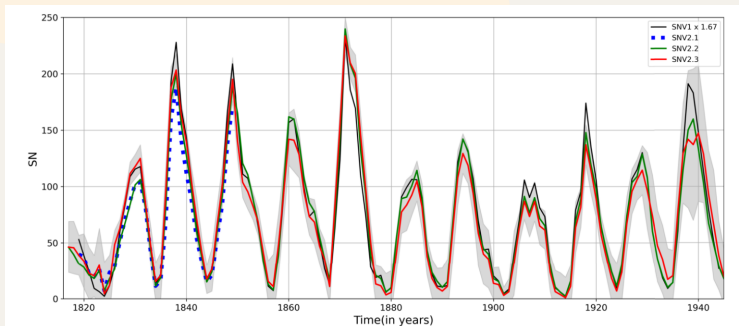
- Dual review of Adams and Gruithuisen counts before ingest.
- Cross-matching against Wolf and SILSO indices to flag drifts.
- Automated scripts check metadata completeness and naming standards.

Toolchain

- Transkribus handwriting models + manual correction interface.
- Python pipelines (Git-tracked) orchestrating OCR, QC, and format exports.
- Planned citizen science portal with training sets for Kurrent transcription.

Data Pipeline to SN V3

1. Curate daily group/spot counts with provenance-rich metadata.
2. Solve observer response curves to harmonise scales over centuries.
3. Aggregate to monthly, yearly, and hemispheric indices with propagated uncertainties.
4. Benchmark against SN V2, satellite-era proxies, and reconstructions to validate corrections.



Historical Assets

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Key Historical Collections

- **Zürich Tables (1945–1979):** 95% digitised; metadata for late-1970s observers underway.
- **Wolf's *Mittheilungen* (1610–1918):** fully integrated with observer statistics over 400+ years.
- **Gruithuisen Manuscripts (1817–1849):** 25% transcribed; bespoke renaming + QC scripts for Kurrent entries.
- **C. H. Adams Drawings (1819–1823):** 338 sketches extracted, 1 056 entries including spotless-day tags.
- **Augustin Stark Ledgers (1813–1835):** harmonising tables + weather annotations.
- **Rost Journals (18th–19th c.):** metadata vocabulary build prior to digitisation ingest.
- **Supporting collections:** Gruithuisen weather logs, Adams correspondence, Zürich observer notes.

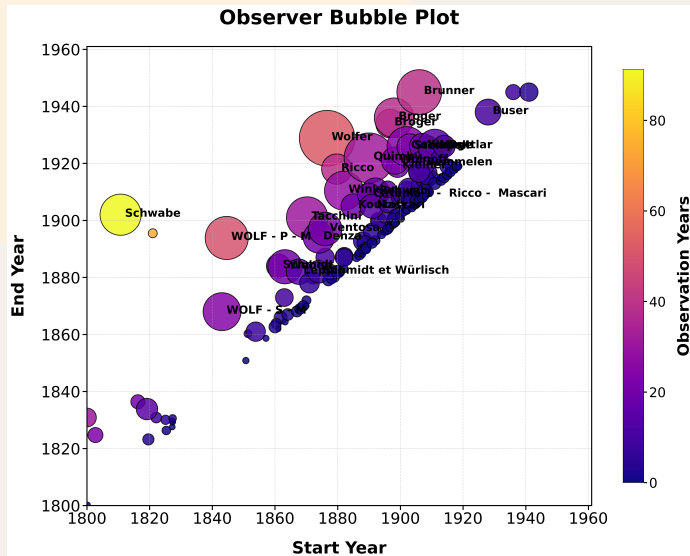


Spotlight: Zürich Observation Tables

- 2 000+ daily tables (1945–1979) digitised with 95% completeness; metadata finished through 1959.
- Capture observer-specific counts, k-factors, weather notes, and substitutions used for the published SN.
- Active work: encode late Waldmeier-era metadata, confirm 1980 holdings, link to USET drawings for positional cross-checks.
- Output feeds the calibration step that harmonises post-war observers with earlier series.

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Observer Coverage



Bubble size = total observations;

Manuscripts and Drawings in Practice

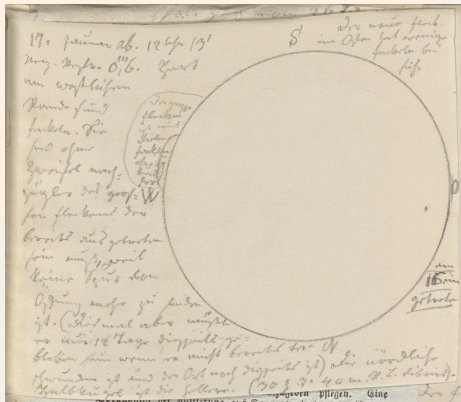


Figure 2: Gruithuisen Kurrent manuscript with weather + spot notes (1819). Dual QC resolves transcription uncertainties.

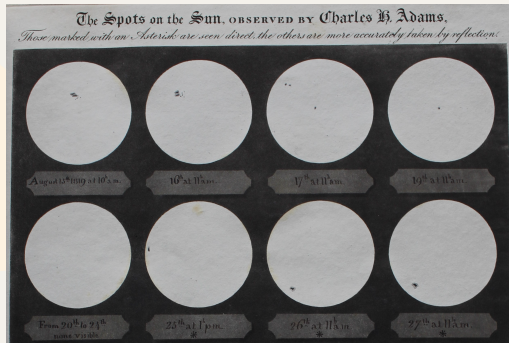


Figure 3: C. H. Adams full-disk drawing (1821). Direct vs. reflection viewing flags underpin spotless-day confidence.

Science Impact

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Addressing Core Challenges

Data hurdles

- Deciphering 19th-century *Kurrent* handwriting and marginalia.
- Normalising heterogeneous formats (drawings, tables, narrative notes).
- Tracking provenance from scanned folio to released dataset.

Mitigations

- Specialist training, handwriting models, and dual QC.
- Schema mapping + metadata vocabularies across archives.
- Git-backed pipelines with audit trails and uncertainty flags.

Benefits for the Community

Forecasting Bias-reduced SN V3 improves solar cycle prediction skill and operational space weather baselines.

Climate Validated long-term forcing records refine Earth system and climate models.

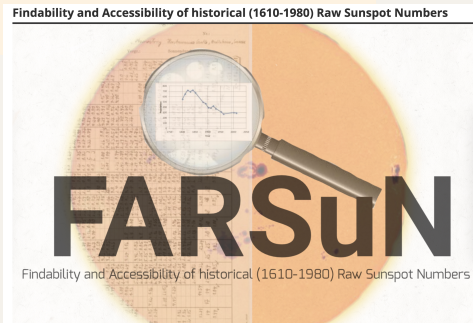
Citizen science Public transcription portal accelerates data recovery and outreach. **Open science** FAIR datasets, APIs, and reproducible notebooks catalyse external research.

Path Ahead

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2025 Roadmap

- Launch citizen-science transcription pilot with curated training sets.
- Finalise Zürich metadata encoding and close remaining digitisation gaps.
- Publish SN V3 release notes with full uncertainty budget and API endpoints.
- Expand collaborations (ROB, Nagoya, UCLouvain) for cross-validation and joint publications.



Q&A

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Questions?

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